

VISHNUTEJA AMBATI

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EDUCATION

Iowa State University

B.S. in Computer Engineering — Transfer Student (2+2 Pathway)

Ames, IA, USA

Aug. 2026 – Expected 2028

BITS Pilani, Hyderabad Campus

B.E. (Hons.) Computer Science — first two years, transferred

Hyderabad, India

Jul. 2024 – May 2026

EXPERIENCE

Founding Engineer

2026 – Present

Hivemind — AI Portfolio Intelligence

Python, LLMs, Knowledge Graphs, RAG, Data Pipelines

- Building *Iris*, an AI portfolio-intelligence platform that maps hidden relationships between an investor's holdings and global financial events—supply chains, ownership, policy, earnings, and macro—across a living knowledge graph of **11,000+ companies** and **27,000+ links**.
- Developing agentic, LLM-powered research pipelines (Ask Iris, Iris Watchlist) that ingest **96+ sources** and **1.2M+ data points/day** into source-cited daily briefs and analyst reports using retrieval-augmented generation (RAG) with traceable citations.

Software Engineering Intern

May 2026 – Jul. 2026

ApexFlo Labs

Python, FastAPI, PostgreSQL, React, TypeScript, PyTorch, ONNX

- Built *Hermathena*, an AI-powered IT Service Management platform that replaces static ticket forms with a natural-language, LLM-driven intake assistant, then routes requests through classification, SLA tracking, and closure over an append-only, event-sourced Python/FastAPI backend on PostgreSQL.
- Shipped a role-based React + TypeScript single-page app (Vite) with server-enforced permissions across four roles, validating every model output before it reached production.
- Built a privacy-first, real-time human-behavior recognition engine running at **18.6 FPS on laptop CPU** (no GPU): fall detection via an ST-GCN (F1 0.78 on URFD) plus rule-based person-down and loitering on one shared backbone, exported to ONNX with 100% prediction agreement.

Software Engineering Intern

Oct. 2025 – Feb. 2026

UzoX (Remote)

Node.js, MongoDB, Docker, Kubernetes

- Engineered backend services for a distributed Discord music platform serving **200K+ monthly active users** and **3M+ weekly audio streams**, owning features end-to-end into production.
- Built RESTful microservices in Node.js (Express) on Docker/Kubernetes with service discovery, and implemented a WebSocket sharding system with caching and load balancing to reach **sub-50ms** real-time latency.

Student Research Assistant — DRDO-Funded Project

Jan. 2026 – May 2026

BITS Pilani, Hyderabad — Computer Vision Surveillance

PyTorch, YOLOv11, CUDA

- Selected for a DRDO (Aeronautics R&D Board) funded project on AI-based human activity detection; built real-time person detection/tracking with YOLOv11 and trained a Temporal Convolutional Network (TCN) on NTU RGB+D 3D-skeleton data (**10K+ sequences**, **24 action classes**) on NVIDIA Tesla T4 GPUs.

RESEARCH & PUBLICATIONS

HAD-Stack: Hamiltonian Stackelberg Game for Robust Fraud Detection

IEEE DSAA 2026

- Co-authored (under review) a physics-guided, game-theoretic framework using quantum-inspired Hamiltonian eigenspectra as a self-regulating controller for adversarial training, removing hand-tuned hyperparameters.
- Reached **0.9948** clean AUC-ROC on 26,393 real UPI transactions while halving false positives vs. a strong Transformer baseline (42→21) and staying robust (AUC 0.9988) under FGSM evasion attack.

PROJECTS

PRISM — Maternal-Health Clinical Decision Support | built for ICMR

React, TypeScript, Supabase

- Developed a full-stack clinical dashboard for a project with the **Indian Council of Medical Research (ICMR)**, India's apex biomedical research body, helping Medical Officers manage high-risk pregnancies via a typed, reusable clinical rules engine with risk explainability and role-based access.

Multimodal Agricultural Intelligence Platform

Python, RAG, LLMs, Multilingual AI, Vector Search, Weather APIs

- Built an AI-powered agricultural intelligence platform integrating **multilingual RAG**, PDF knowledge indexing, semantic retrieval, and weather-aware services to generate context-aware farming assistance from domain-specific agricultural data.

Smart Hospital Edge AI on FPGA

PyTorch, Verilog, Zynq-7020, INT8

- Deployed a real-time ECG anomaly detector as Verilog RTL on a Xilinx Zynq-7020; trained a 1D CNN on MIT-BIH with a custom PyTorch→**INT8 quantization** pipeline (4× smaller) and a deterministic FSM-based MAC inference engine at 100 MHz.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, TypeScript, Go, Rust, SQL, Verilog

Distributed & Backend Systems: Distributed Systems, Real-Time Systems, Microservices, WebSockets, REST APIs, Node.js, Express, FastAPI, Docker, Kubernetes, PostgreSQL, MongoDB, Supabase, Event-Sourcing, Caching, Load Balancing, System Design, CI/CD

AI / ML: PyTorch, TensorFlow, ONNX, Scikit-learn, OpenCV, PyTorch Geometric, Graph Neural Networks, CNNs, TCNs, ST-GCN, Computer Vision, Adversarial ML, LLMs, RAG, Agentic AI, Knowledge Graphs, Prompt Engineering, Quantization

Data, Cloud & Foundations: Data Pipelines/ETL, Feature Engineering, Prometheus, Grafana, AWS, Google Cloud, Git, Linux, CUDA (Tesla T4); Data Structures & Algorithms, OOP, DBMS, Concurrency